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A Mathematical Framework for Psychophysical Interaction: Discrete Foundations, Continuous Extension, and Behavioral Applications --Manuscript Draft--

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Abstract:	We develop a mathematical framework for representing psychophysical interaction dynamics between agents, distinguishing between active (expressive) and passive (perceptual) interaction states. By introducing an explicit interaction-exposure operator A alongside a psychophysical-response matrix D, we formulate the interaction operator $\gamma = DA$. This factorization removes an implicit rank-one limitation on γ and isolates self-coupling from cross-agent coupling. Under a local logarithmic approximation, we formalize the discrete-time system $I_{S,k+1} = \gamma I_{S,k}$ and establish its existence, uniqueness, fixed points, spectral stability, second-order Jury conditions, and oscillatory regimes. We extend the model to continuous time via a semigroup representation $\Gamma(t) = e^{tM}$, characterizing the exact conditions under which a discrete system admits a real continuous generator. Furthermore, we show that the locally linearized Gottman dyadic model of marital interaction embeds structurally within this framework, and the Thorndike exponential learning curve embeds under a sign-determined constitutive flow assumption. The framework provides a rigorous foundation for integrating psychophysical encoding with interpersonal dynamic systems.

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Editor-in-Chief

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Subject: Submission of manuscript: “A Mathematical Framework for Psychophysical Interaction: Discrete Foundations, Continuous Extension, and Behavioral Applications”

Dear Editor,

I am pleased to submit our manuscript titled “**A Mathematical Framework for Psychophysical Interaction: Discrete Foundations, Continuous Extension, and Behavioral Applications**” for consideration for publication as an Original Research Article in the *Journal of Mathematical Psychology*.

This work addresses a theoretical gap at the intersection of psychophysics and dynamical systems modeling of interpersonal interaction. While classical psychophysics models stimulus-percept relations and dynamic models describe interaction trajectories, a unified framework connecting psychophysical scaling directly to interaction operators has remained undeveloped.

In this paper, we construct a mathematically rigorous linear framework that separates psychophysical sensitivity from cross-agent interaction exposure ($\gamma = DA$). This structural factorization overcomes the rank-one restriction inherent in scalar psychophysical compositions, allowing complex spectral and oscillatory behavior. We provide complete discrete stability criteria (via second-order Jury conditions), establish continuous-time matrix semigroup extensions, and derive the necessary conditions for real generator existence. Crucially, we demonstrate the utility of this theoretical foundation by proving that established empirical models—specifically the locally linearized Gottman dyadic interaction model and Thorndike’s exponential learning curve—embed structurally within our framework.

Given the *Journal of Mathematical Psychology*’s strong emphasis on rigorous mathematical formulations of psychological phenomena and formal dynamic models, we believe our manuscript is well-suited for your readership.

We confirm that:

1. This manuscript is original, has not been published previously, and is not currently under consideration for publication elsewhere.
2. The author has no conflicts of interest to declare.

Thank you very much for considering our manuscript. We look forward to hearing from you regarding the peer-review process.

Sincerely,

Chewin Pinmook

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Highlights

- Formulates a mathematical framework coupling psychophysical encoding with agent interaction dynamics.
- Factorizes the interaction operator ($\gamma = DA$) to enable multi-dimensional spectral behavior and overcome rank-one constraints.
- Derives closed-form discrete Jury stability conditions and exact continuous-time semigroup representations.
- Proves structural embedding of the locally linearized Gottman dyadic interaction model.
- Embeds Thorndike's exponential learning curve under an explicit, sign-determined constitutive flow law.

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We develop a mathematical framework for representing psychophysical interaction dynamics between agents, distinguishing between active (expressive) and passive (perceptual) interaction states. By introducing an explicit interaction-exposure operator A alongside a psychophysical-response matrix D , we formulate the interaction operator $\gamma = DA$. This factorization removes an implicit rank-one limitation on γ and isolates self-coupling from cross-agent coupling. Under a local logarithmic approximation, we formalize the discrete-time system $I_{S,k+1} = \gamma I_{S,k}$ and establish its existence, uniqueness, fixed points, spectral stability, second-order Jury conditions, and oscillatory regimes. We extend the model to continuous time via a semigroup representation $\Gamma(t) = e^{tM}$, characterizing the exact conditions under which a discrete system admits a real continuous generator. Furthermore, we show that the locally linearized Gottman dyadic model of marital interaction embeds structurally within this framework, and the Thorndike exponential learning curve embeds under a sign-determined constitutive flow assumption. The framework provides a rigorous foundation for integrating psychophysical encoding with interpersonal dynamic systems.

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Abstract

We develop a mathematical framework for representing psychophysical interaction dynamics between agents. The framework distinguishes an active (expressive) interaction state from a passive (perceptual) state and introduces an explicit interaction-exposure operator that separates psychophysical sensitivity from cross-agent coupling. Under a local logarithmic psychophysical approximation and a linear coupling assumption, the model yields a discrete-time linear system $I_{S,k+1} = \gamma I_{S,k}$, $\gamma = DA$, where D is a diagonal psychophysical-response matrix and A is an interaction-exposure matrix. This factorization removes an otherwise implicit rank-one restriction on γ and gives an independent interpretation to self- and cross-agent coupling coefficients. We characterize existence and uniqueness, fixed points, spectral stability, the second-order Jury conditions, and oscillatory regimes for the discrete system, and derive the corresponding continuous-time semigroup representation $\Gamma(t) = e^{tM}$ together with the precise conditions under which a discrete matrix admits such a continuous generator. We distinguish spectral crossings from genuine nonlinear Hopf or Neimark–Sacker bifurcations, which require nonlinear dynamics not claimed here. We show that the locally linearized Gottman dyadic model of marital interaction embeds structurally in the proposed linear framework, and that the Thorndike exponential learning curve embeds under an explicit, sign-determined constitutive flow assumption. A worked numerical example illustrates the discrete stability criteria on a concrete parameterization. The framework is explicitly theoretical: it es-

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establishes internal mathematical consistency and representational scope, not empirical universality, and we identify parameter inference and empirical validation as open problems for future work.

Keywords: psychophysics, Weber–Fechner law, dyadic interaction, linear dynamical systems, Gottman model, matrix semigroups, stability theory

1. Introduction

1.1. *Motivation and Context*

Interpersonal interaction can be represented as a dynamical process in which expressed signals alter the perceptual states of other agents and thereby influence subsequent expression. A mathematical treatment therefore requires a joint representation of perception–action coupling and a temporal evolution law. Psychophysics provides one possible local model for the relation between stimulus magnitude and perceived intensity.

The classical Weber–Fechner relation is logarithmic over an appropriate physical stimulus range. Recent work has investigated Weber-type scaling in affective and value-related settings [7, 1]. In particular, Berkovich and Meiran [1] provide direct evidence, using evidence-accumulation modeling of pleasantness judgments, that self-reported pleasant affect obeys Weber’s law over the intensity range they tested. The present paper does not assume that such scaling holds universally for interpersonal behavior; it adopts logarithmic scaling as a restricted modeling assumption over a moderate-intensity regime, consistent with the scope in which it has been empirically observed.

1.2. *Scope and Intent*

This work is theoretical. Its purpose is to construct a mathematically coherent representation in which psychophysical encoding, interaction coupling, and linear dynamical evolution can be studied within a single formal structure, and to establish exactly which existing dyadic models (Section 7) and learning phenomena (Section 8) are representable within that structure. The model is not offered as a complete empirical theory of interpersonal behavior. Nonlinear saturation, memory, stochasticity, heterogeneous time scales, and empirical parameter estimation from data are outside the present linear formulation; Section 11 outlines how each of these extensions attaches to the framework developed here.

1.3. Contribution Relative to Existing Models

Two existing lines of work are the natural comparison points. The Gottman dyadic model [3] specifies a nonlinear difference equation for marital interaction and has an established locally linearized regime; the present framework does not compete with that empirical model but instead asks a structural question about it: under what general class of linear interaction operators does its linearization sit, and what does that class allow beyond the two-agent case that motivated it? Proposition 7.1 answers this by exhibiting the exact matrix identification, and the surrounding sections show that the identification is not an isolated coincidence but an instance of the general representation $I_P = DAI_S$.

The distinguishing technical content of this paper, relative to treating γ as an unstructured coupling matrix (as in a direct linearization of a dyadic model), is threefold: (i) the factorization $\gamma = DA$ in Theorem 3.3, which shows that entrywise psychophysical scaling alone would force $\text{rank}(\gamma) \leq 1$ (Remark 3.4), and that the interaction-exposure operator A is the minimal additional structure needed to escape that restriction; (ii) the exact continuous-time correspondence in Theorem 5.3 together with the converse obstruction in Remark 5.5, which is often stated informally as “discrete-to-continuous is just $\ln \gamma / \Delta t$ ” without noting that a real logarithm need not exist; and (iii) the complete second-order Jury characterization (Theorem 4.4) applied to the dyadic case, which gives necessary and sufficient stability conditions in closed form rather than the determinant condition alone. Section 4.8 gives a numerical instance of all three.

1.4. Contributions

The formulation makes the following contributions:

- (1) It separates psychophysical sensitivity from interaction exposure through the factorization $\gamma = DA$.
- (2) It provides a discrete-time linear interaction system and its exact stability conditions (existence/uniqueness, fixed points, spectral stability, Jury criteria).
- (3) It identifies the precise additional semigroup assumption required for the continuous representation $\gamma(t) = e^{tM}$, and states the converse obstruction (non-existence of a real generator) explicitly.
- (4) It characterizes oscillatory regimes through the spectrum while explicitly distinguishing spectral crossings from genuine nonlinear bifurcations.

- (5) It gives a structural representation of the locally linearized Gottman model without claiming that the latter is empirically derived from psychophysics.
- (6) It embeds the Thorndike exponential learning curve under an explicit, sign-determined flow assumption (Section 8), rather than a sign chosen post hoc to match the target curve.

2. Psychophysical Foundations

2.1. The Weber–Fechner Law

For a positive stimulus magnitude S and reference level S_0 , the classical logarithmic relation is

$$P = \alpha \ln\left(\frac{S}{S_0}\right), \quad \alpha > 0. \quad (1)$$

It follows formally from the differential relation $dP = \alpha \frac{dS}{S}$. The logarithmic form implies diminishing marginal changes in perceived intensity as stimulus magnitude increases.

2.2. Restricted Generalization to Social-Emotional Stimuli

Assumption 2.1 (Restricted psychophysical generalization). Over a specified moderate-intensity regime, a social-emotional stimulus may be represented locally by

$$P_{\text{social}} \approx \alpha \ln\left(\frac{S_{\text{social}}}{S_0}\right). \quad (2)$$

This is a modeling assumption, not a claim of universal psychological validity.

Assumption 2.1 is compatible with the existence of Weber-type effects in affective and value-related measurements documented by Namboodiri et al. [7] and Berkovich and Meiran [1], but empirical validation for the specific interaction variables I_S, I_P introduced below is not supplied in this paper and is identified as future work in Section 11. However, these variables (I_S, I_P) can be empirically operationalized using measurement protocols established in Gottman’s marital experiments, such as SPAFF (Specific Affect Coding System) scores and Continuous Affect Scores derived from rating dials[3] [15] [16].

2.3. Perception–Action Feedback

The conceptual structure is

$$S \longrightarrow I_S \longrightarrow I_P \longrightarrow P,$$

with feedback through repeated interaction. The central mathematical assumption is that, after psychophysical encoding, the relevant interaction variables admit a local linear coupling representation.

3. Mathematical Objects and Core Definitions

3.1. Interaction Variables

Definition 3.1 (Active interaction). For n agents, the active interaction state is $I_{S,k} \in \mathbb{R}^n$, where $[I_{S,k}]_i$ denotes the expressive interaction signal of agent i at interaction step k .

Definition 3.2 (Passive interaction). The passive interaction state is $I_{P,k} \in \mathbb{R}^n$, where $[I_{P,k}]_i$ denotes the encoded perceptual response of agent i to the interaction exposure it receives.

3.2. Psychophysical Encoding

Let $S_i > 0$ denote the effective stimulus magnitude received by agent i , and let x_i denote the corresponding effective stimulus exposure of agent i after interaction transmission (defined formally in Section 3.3). Define

$$[I_P]_i = k_{P,i} \alpha_i \ln \left(\frac{x_i}{S_0} \right), \quad (3)$$

$$[I_S]_i = k_{S,i} \ln \left(\frac{S_i}{S_0} \right), \quad (4)$$

where $k_{P,i}, k_{S,i} > 0$ are agent-specific scaling coefficients. Let

$$D = \text{diag} \left(\frac{k_{P,1} \alpha_1}{k_{S,1}}, \dots, \frac{k_{P,n} \alpha_n}{k_{S,n}} \right).$$

Then the direct psychophysical mapping is

$$I_P = Dx. \quad (5)$$

3.3. Interaction-Exposure Operator

To represent cross-agent transmission, introduce a dimensionless interaction-exposure matrix $A \in \mathbb{R}^{n \times n}$. Its element A_{ij} measures the local contribution of active signal j to the effective stimulus exposure received by agent i . The effective encoded stimulus vector $x = (x_1, \dots, x_n)^\top$ from Section 3.2 is modeled locally by

$$x = AI_S. \quad (6)$$

The corresponding passive state is

$$I_P = DAI_S. \quad (7)$$

Theorem 3.3 (Interaction operator representation). *Under the local linear exposure assumption $x = AI_S$, the passive and active states satisfy*

$$I_P = \gamma I_S, \quad \gamma := DA. \quad (8)$$

Moreover, $\gamma_{ij} = D_{ii}A_{ij}$, so diagonal terms represent self-coupling and off-diagonal terms represent cross-agent coupling.

Proof. By definition, $I_P = Dx$ and $x = AI_S$. Hence $I_P = DAI_S$. Setting $\gamma = DA$ gives the stated representation. Since D is diagonal, $\gamma_{ij} = D_{ii}A_{ij}$. $\square$

Remark 3.4 (Necessity of the exposure operator). The factorization $\gamma = DA$ is not a notational convenience; it is load-bearing. If interaction were instead modeled by direct entrywise composition of scalar psychophysical sensitivities, $\gamma_{ij} = \frac{k_{P,i}\alpha_i}{k_{S,j}}$, then γ would factor as an outer product $uv^\top$ with $u_i = k_{P,i}\alpha_i$ and $v_j = \frac{1}{k_{S,j}}$, forcing $\text{rank}(\gamma) \leq 1$ regardless of n . A rank-one γ cannot support the two-dimensional eigenvalue structure needed for the oscillatory regime of Section 4.7, since a rank-one real matrix has at most one nonzero eigenvalue. The independent exposure matrix A is therefore the minimal structural addition required for the framework to admit oscillatory or complex-conjugate spectral behavior at all.

4. Discrete-Time Interaction Dynamics

4.1. Discrete Interaction System

We impose the following local closure: the passive interaction response at one interaction step determines the active interaction state at the next step. Thus

$$I_{S,k+1} = \gamma I_{S,k}, \quad k \in \mathbb{N}. \quad (9)$$

This is a model of sampled perception–action cycles; it does not assert that interpersonal interaction is intrinsically discrete.

4.2. Existence and Uniqueness

Theorem 4.1 (Existence and uniqueness of discrete evolution). *For every $I_{S,0} \in \mathbb{R}^n$ and fixed $\gamma \in \mathbb{R}^{n \times n}$, the recurrence $I_{S,k+1} = \gamma I_{S,k}$ has the unique solution*

$$I_{S,k} = \gamma^k I_{S,0}. \quad (10)$$

Proof. Existence follows by induction. For $k = 0$ the formula is the initial condition. If $I_{S,k} = \gamma^k I_{S,0}$, then $I_{S,k+1} = \gamma I_{S,k} = \gamma^{k+1} I_{S,0}$. Uniqueness follows because each state is uniquely determined by the preceding state via (9). $\square$

4.3. Equilibria

Lemma 4.2 (Fixed-point structure). *A state I_S^* is an equilibrium if and only if*

$$(\gamma - I)I_S^* = 0. \quad (11)$$

Thus the equilibrium set is $\ker(\gamma - I)$, and the origin is the unique equilibrium if and only if 1 is not an eigenvalue of γ .

4.4. Time-Varying Interactions

For a time-varying coupling matrix,

$$I_{S,k+1} = \gamma_k I_{S,k}, \quad (12)$$

with solution

$$I_{S,k} = \gamma_{k-1} \cdots \gamma_1 \gamma_0 I_{S,0}. \quad (13)$$

This extension can represent adaptation, learning, fatigue, or context dependence, but stability then requires analysis of the matrix product in (13) rather than only the spectrum of a single fixed matrix; we do not develop joint-spectral-radius conditions here and flag this as future work (Section 11).

4.5. Spectral Stability

Proposition 4.3 (Asymptotic stability). *The origin of the fixed-matrix system (9) is asymptotically stable if and only if*

$$\rho(\gamma) < 1, \quad (14)$$

where $\rho(\gamma) = \max_{\lambda \in \text{spec}(\gamma)} |\lambda|$.

Proof. This is the standard spectral characterization of asymptotic stability for a finite-dimensional linear discrete system. Jordan blocks contribute at most polynomial factors in k multiplied by $|\lambda|^k$, which converge to zero exactly when every eigenvalue satisfies $|\lambda| < 1$. $\square$

4.6. Dyadic-Agent Specialization

For two agents,

$$\gamma = \begin{pmatrix} \gamma_{AA} & \gamma_{AB} \\ \gamma_{BA} & \gamma_{BB} \end{pmatrix}. \quad (15)$$

Its characteristic polynomial is

$$p(\lambda) = \lambda^2 - \text{tr}(\gamma)\lambda + \det(\gamma), \quad (16)$$

with roots

$$\lambda_{\pm} = \frac{\text{tr}(\gamma) \pm \sqrt{\text{tr}(\gamma)^2 - 4\det(\gamma)}}{2}. \quad (17)$$

Theorem 4.4 (Jury stability criterion). *The dyadic system is asymptotically stable if and only if*

$$|\det(\gamma)| < 1, \quad (18)$$

$$1 + \text{tr}(\gamma) + \det(\gamma) > 0, \quad (19)$$

$$1 - \text{tr}(\gamma) + \det(\gamma) > 0. \quad (20)$$

Proof. For $p(\lambda) = \lambda^2 + a_1\lambda + a_0$, the second-order Jury criterion is $|a_0| < 1$, $1 + a_1 + a_0 > 0$, and $1 - a_1 + a_0 > 0$. Here $a_1 = -\text{tr}(\gamma)$ and $a_0 = \det(\gamma)$, giving (18)–(20). $\square$

Remark 4.5. The determinant condition (18) alone is necessary but not sufficient for asymptotic stability; all three conditions (18)–(20) are required jointly. This is used in Section 7 to caveat the relationship between $\det(\gamma)$ and stability of the linearized Gottman system.

4.7. Oscillatory Regime

Define

$$\Delta_\gamma = \text{tr}(\gamma)^2 - 4 \det(\gamma) = (\gamma_{AA} - \gamma_{BB})^2 + 4\gamma_{AB}\gamma_{BA}. \quad (21)$$

Proposition 4.6 (Complex-eigenvalue criterion). *The dyadic discrete system has a complex-conjugate eigenvalue pair if and only if*

$$\Delta_\gamma < 0, \quad (22)$$

equivalently

$$-\gamma_{AB}\gamma_{BA} > \frac{(\gamma_{AA} - \gamma_{BB})^2}{4}. \quad (23)$$

This condition identifies an oscillatory spectral regime. It does not by itself establish a psychological or clinical interpretation of the oscillation; see Section 9.

4.8. Worked Numerical Example

To make the discrete criteria concrete, consider a symmetric-exposure dyad with

$$D = \text{diag}(0.9, 0.7), \quad A = \begin{pmatrix} 0.30 & 0.55 \\ 0.40 & 0.35 \end{pmatrix},$$

so that

$$\gamma = DA = \begin{pmatrix} 0.270 & 0.495 \\ 0.280 & 0.245 \end{pmatrix}.$$

Then $\text{tr}(\gamma) = 0.515$ and $\det(\gamma) = (0.270)(0.245) - (0.495)(0.280) = 0.06615 - 0.1386 = -0.07245$. Checking the Jury conditions (18)–(20): $|\det(\gamma)| = 0.07245 < 1$; $1 + \text{tr}(\gamma) + \det(\gamma) = 1.4426 > 0$; $1 - \text{tr}(\gamma) + \det(\gamma) = 0.4126 > 0$. All three hold, so by Theorem 4.4 the origin is asymptotically stable. The discriminant is $\Delta_\gamma = 0.515^2 - 4(-0.07245) = 0.2652 + 0.2898 = 0.5550 > 0$, so by Proposition 4.6 the eigenvalues are real (no oscillatory component): explicitly $\lambda_\pm = (0.515 \pm \sqrt{0.5550})/2$, giving $\lambda_+ \approx 0.630$ and $\lambda_- \approx -0.115$, both inside the unit disk, consistent with the Jury result. This example also illustrates Remark 3.4: γ here has rank 2, which is unreachable under an entrywise (rank-one) coupling assumption, and its stabilization depends jointly on D and A rather than on either factor alone.

4.9. Spectral Crossings

A parameter value μ_c is called a *spectral crossing point* when an eigenvalue of $\gamma(\mu)$ reaches the unit circle. Such a crossing indicates a change in linear stability. A nonlinear Neimark–Sacker bifurcation requires additional nonlinear terms and nondegeneracy hypotheses and is not claimed here.

5. Continuous-Time Extension

5.1. Interaction Operator Family

Let $\Gamma(t)$ denote a family of interaction operators for elapsed time $t \geq 0$.

Assumption 5.1 (Continuous semigroup). The family satisfies

$$\Gamma(0) = I, \quad \Gamma(t + s) = \Gamma(t)\Gamma(s), \quad (24)$$

and is continuous in t .

Definition 5.2 (Interaction-flow generator). If the derivative at zero exists, define

$$M = \left. \frac{d}{dt} \Gamma(t) \right|_{t=0}. \quad (25)$$

Theorem 5.3 (Generator representation). *Under the continuous-semigroup Assumption 5.1 in finite-dimensional state space,*

$$\Gamma(t) = e^{tM} \quad (26)$$

for all $t \geq 0$.

Proof. In finite dimensions every continuous matrix semigroup is norm-continuous and hence generated by a constant matrix M [9]. Equivalently, differentiating the semigroup identity gives $\Gamma'(t) = M\Gamma(t)$ with $\Gamma(0) = I$. The unique solution of this linear ODE is $\Gamma(t) = e^{tM}$. $\square$

Corollary 5.4 (Discrete sampling). *For a sampling interval $\Delta t > 0$,*

$$\gamma = \Gamma(\Delta t) = e^{M\Delta t}. \quad (27)$$

Thus the discrete and continuous descriptions are exactly consistent when the discrete operator belongs to a continuous semigroup.

Remark 5.5 (Converse and matrix logarithms). Given an arbitrary discrete matrix γ , a continuous-time real generator M satisfying $\gamma = e^{M\Delta t}$ need not exist. If such an M exists, it is a real matrix logarithm of γ divided by Δt . A sufficient condition for the principal logarithm to be real is that γ have no eigenvalues on the closed negative real axis [10]. Invertibility of γ alone does not guarantee a real logarithm: a real 2×2 matrix with a negative real eigenvalue of odd algebraic multiplicity structure can be invertible yet admit no real logarithm. Consequently, fitting a discrete γ from data does not automatically license a continuous-time reinterpretation, and this step should be checked explicitly rather than assumed.

5.2. Continuous Dynamics

The continuous interaction state satisfies

$$\frac{dI_S}{dt} = MI_S, \quad (28)$$

with solution

$$I_S(t) = e^{Mt} I_S(0). \quad (29)$$

6. Spectral Theory and Qualitative Regimes

6.1. Continuous-Time Stability

Theorem 6.1 (Continuous stability criterion). *The origin of $\dot{I}_S = MI_S$ is asymptotically stable if and only if*

$$\operatorname{Re} \lambda < 0 \quad (30)$$

for every $\lambda \in \operatorname{spec}(M)$.

Proof. The eigenvalues of e^{Mt} are $e^{\lambda t}$. Their magnitudes decay to zero exactly when $\operatorname{Re} \lambda < 0$ for every eigenvalue, including the standard Jordan-block polynomial factors. $\square$

6.2. Two-Agent Oscillation

For

$$M = \begin{pmatrix} m_{AA} & m_{AB} \\ m_{BA} & m_{BB} \end{pmatrix},$$

the discriminant is

$$\Delta_M = (m_{AA} - m_{BB})^2 + 4m_{AB}m_{BA}. \quad (31)$$

Complex eigenvalues occur exactly when $\Delta_M < 0$, equivalently

$$-m_{AB}m_{BA} > \frac{(m_{AA} - m_{BB})^2}{4}. \quad (32)$$

In that regime, trajectories contain oscillatory components. The interpretation of such oscillation as, for example, affective cycling is a model hypothesis to be tested empirically, not a mathematical theorem.

6.3. Qualitative Regimes

Regime	Spectral condition	Interpretation
Convergence	$\text{Re } \lambda < 0$	exponential decay
Divergence	$\text{Re } \lambda > 0$	exponential growth
Neutral	$\text{Re } \lambda = 0$	marginal behavior
Oscillatory	$\text{Im } \lambda \neq 0$	oscillatory component
Oscillatory instability	$\text{Re } \lambda > 0, \text{ Im } \lambda \neq 0$	growing oscillation

Table 1: Qualitative regimes of the linear continuous-time system, classified by the spectrum of M .

6.4. Spectral Crossing versus Hopf Bifurcation

Let $M(\mu)$ depend smoothly on a parameter μ . If a conjugate eigenvalue pair satisfies

$$\lambda_{\pm}(\mu_c) = \pm i\omega, \quad \omega \neq 0,$$

and crosses the imaginary axis transversely,

$$\frac{d}{d\mu} \text{Re } \lambda_{\pm}(\mu_c) \neq 0,$$

then the linearization undergoes a Hopf-type *spectral crossing*. This indicates a change from decay to neutral oscillation in the linearized system only. A genuine nonlinear Hopf bifurcation, including the birth of a periodic orbit, requires nonlinear terms and additional nondegeneracy conditions [14] and is reserved for future work.

7. Relation to the Gottman Dyadic Model

7.1. Local Linearization

Consider the dyadic equations of Gottman et al. [3]:

$$W_{k+1} = a + r_1 W_k + I_{HW}(H_k), \quad (33)$$

$$H_{k+1} = b + r_2 H_k + I_{WH}(W_k). \quad (34)$$

After shifting to an equilibrium and locally linearizing the influence functions,

$$I_{HW}(H) \approx c_{HW}H, \quad I_{WH}(W) \approx c_{WH}W,$$

we obtain

$$y_{k+1} = G y_k, \quad G = \begin{pmatrix} r_1 & c_{HW} \\ c_{WH} & r_2 \end{pmatrix}. \quad (35)$$

Proposition 7.1 (Structural representation). *The locally linearized homogeneous Gottman system has exactly the same mathematical form as the proposed interaction system (9) under the identification*

$$\gamma_{AA} = r_1, \quad \gamma_{AB} = c_{HW}, \quad \gamma_{BA} = c_{WH}, \quad \gamma_{BB} = r_2.$$

Theorem 7.2 (Stability equivalence under matrix identification). *Under the identification of Proposition 7.1, the linearized Gottman system is asymptotically stable if and only if $\rho(\gamma) < 1$.*

Proof. The two systems are governed by the same homogeneous matrix under the stated identification. Therefore their trajectories have identical spectral stability properties by Proposition 4.3. $\square$

Remark 7.3. This is a structural dynamical equivalence, not a derivation of Gottman’s empirical model from Weber–Fechner psychophysics (see Section 1.3). The present framework provides a mathematical representation in which the local linearized Gottman system embeds; the psychophysical encoding of Section 2 supplies one possible, but not the only, microfoundation for the coefficients c_{HW}, c_{WH} .

The determinant identity $\det(\gamma) = r_1 r_2 - c_{HW} c_{WH}$ is useful diagnostically, but by Theorem 4.4 the determinant condition alone is only one of three jointly necessary and sufficient conditions; no equivalence between a single determinant inequality and full Jury stability should be inferred without checking (19)–(20) as well.

If γ is known to be generated by a continuous semigroup (Assumption 5.1), then $\gamma = e^{M\Delta t}$, and the corresponding continuous generator M (subject to the existence caveat of Remark 5.5) provides a continuous-time representation of the same linearized dynamics.

8. Behavioral Application: The Thorndike Learning Curve

The purpose of this section is to provide an embedding of a known empirical curve within the present framework, not a unique derivation of it from first principles.

Let $t_S(n)$ denote escape time after trial n , and let I_S be a scalar performance variable, with the sign convention that I_S increases with practice (higher performance). Escape time is inversely related to performance, so an increasing I_S corresponds to a *decreasing* t_S . We therefore adopt, a priori and not as a fitting choice,

$$\frac{dI_S}{dt} = +AI_S, \quad A > 0, \quad (36)$$

reflecting that performance is reinforced by successful trials, and represent the relation between performance and escape time through a constitutive coupling $M(t_S)$,

$$\frac{dI_S}{dt_S} = -M(t_S)I_S, \quad (37)$$

where the negative sign is fixed by the requirement that t_S decreases as I_S increases along a trajectory (i.e. $dt_S/dn < 0$ whenever $dI_S/dn > 0$ and $M(t_S) > 0$); this is a statement about the direction of the I_S - t_S relationship, not a free parameter chosen after the fact.

By the chain rule,

$$\frac{dI_S}{dn} = -M(t_S)I_S \frac{dt_S}{dn}. \quad (38)$$

Equating (36) (rewritten as $dI_S/dn = AI_S$, treating n as the flow parameter) with (38) and assuming $M(t_S) \neq 0$ gives

$$\frac{dt_S}{dn} = -\frac{A}{M(t_S)}. \quad (39)$$

The negative sign here is inherited directly from (37) and is therefore determined by the modeling choice made before any curve-fitting, not selected to reproduce a decreasing curve after the fact.

For the constitutive choice $M(t_S) = c/t_S$ with $c > 0$ (i.e., the coupling strengthens as escape time shrinks), Equation (39) becomes

$$\frac{dt_S}{dn} = -\frac{A}{c} t_S, \quad (40)$$

with solution

$$t_S(n) = t_S(0) \exp\left(-\frac{A}{c} n\right). \quad (41)$$

Thus the exponential learning curve is an admissible embedding of the interaction-flow framework under the specified constitutive law $M(t_S) = c/t_S$ and the sign convention fixed in (36)–(37); a different constitutive choice for $M(t_S)$ would yield a different learning curve within the same framework, and Equation (41) is not claimed to be uniquely implied by the general theory of Sections 4–6.

Figure 1 compares the deterministic solution (41) with a stochastic version obtained by adding proportional trial-and-error noise, $dt_S = -(A/c) t_S dn + \sigma t_S dW_n$, simulated by the Euler–Maruyama method with $\sigma = 0.35$.

9. Discussion

The framework separates three logically distinct ingredients: psychophysical encoding (Section 2), interaction coupling (Section 3.3), and temporal dynamics (Sections 4–6). The psychophysical component supplies a local scaling map. The interaction-exposure matrix supplies cross-agent structure, and Remark 3.4 shows this structure is not optional if oscillatory dynamics are to be representable at all. The matrix γ , or its continuous generator M , then governs the linear state evolution.

The spectral results provide mathematically precise statements. In discrete time, $\rho(\gamma) < 1$ gives asymptotic stability (Proposition 4.3); the full Jury system (18)–(20) gives this in closed form for the dyadic case, as illustrated numerically in Section 4.8. In continuous time, $\text{Re } \lambda(M) < 0$ for every eigenvalue gives asymptotic stability. Complex eigenvalues imply oscillatory components, while positive real parts imply exponential instability.

These mathematical facts should not be conflated with empirical psychological conclusions. Terms such as *emotional resonance*, *escalation*, or *relational cycling* are interpretations of model regimes that require independent observational validation; they are not established by the spectral results themselves. In particular, a positive eigenvalue does not by itself establish

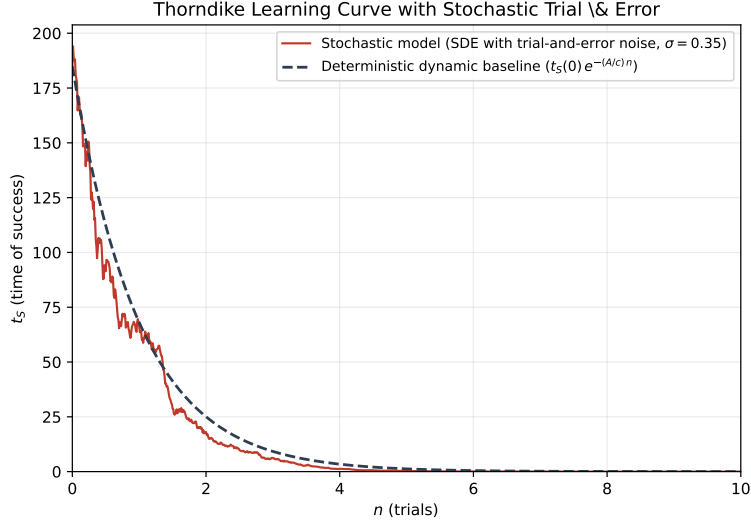


Figure 1: The introduction of state-dependent stochastic noise ($\sigma t_S dW_n$) captures the non-monotonic character of trial-and-error behavior observed in Thorndike’s puzzle-box experiments [8]. Transient spikes during early trials ($n < 4$) reflect exploratory behavior, while the attenuation of fluctuations as $t_S \rightarrow 0$ corresponds to habit formation and skill mastery. Both curves use the decay rate A/c from Equation (41), illustrated here with $A/c = 1$; the deterministic curve is the $\sigma \rightarrow 0$ limit of the stochastic model.

that a modeled relationship is dysfunctional, and an oscillatory eigenvalue pair does not by itself establish an emotional-resonance mechanism. The framework does not claim to explain subjective emotional experience, narrative cognition, or phenomenological meaning: the variables I_S and I_P are abstract interactional states, and different psychophysical laws could be substituted for the logarithmic law of Section 2 while preserving the same dynamical architecture of Sections 3–6.

10. Conclusion

We have formulated a mathematical framework for psychophysical interaction based on a local logarithmic encoding assumption, an explicit interaction-exposure operator, and linear state dynamics. The key representation is

$$I_P = DAI_S = \gamma I_S, \quad I_{S,k+1} = \gamma I_{S,k}.$$

Under a continuous-semigroup assumption, the continuous counterpart is

$$\frac{dI_S}{dt} = MI_S, \quad \gamma = e^{M\Delta t}.$$

The framework yields stability and spectral results, including the full Jury criterion for the dyadic discrete system (illustrated numerically in Section 4.8) and real-part stability criteria for continuous time. Complex eigenvalues identify oscillatory components, while spectral crossings identify changes in linear stability; genuine nonlinear Hopf or Neimark–Sacker bifurcations are not claimed in this linear theory.

The locally linearized Gottman model represents structurally within the framework (Proposition 7.1), and the Thorndike exponential learning curve embeds under an explicit, sign-determined constitutive flow assumption (Section 8). These results demonstrate mathematical compatibility with two independently established models, rather than empirical derivation of either from psychophysics or a claim of psychological universality for the framework itself.

11. Future Work

Three directions are especially important for extending this theoretical framework toward empirical use:

- (1) *Nonlinear extension.* Introduce $\dot{I}_S = f(I_S; \mu)$ to study saturation, thresholds, memory, genuine Hopf bifurcations, multiple equilibria, and nonlinear attractors, building on the linear spectral-crossing analysis of Section 6.4.
- (2) *Joint-spectral-radius stability for the time-varying case.* The time-varying system of Section 4.4 requires stability conditions on the matrix product (13) rather than the spectrum of a fixed γ ; this is left open here.
- (3) *Empirical parameter inference.* Estimate A , D , γ , or M from dyadic time-series data and test whether the predicted spectral regimes correspond to observed interaction patterns, subject to the real-logarithm existence check of Remark 5.5 whenever a continuous-time reinterpretation is desired.

Declaration of Interest

During the preparation of this work, the author used **ChatGPT**, **Gemini**, **Claude** in order to improve language readability, refine sentence struc-

tures, and assist in formatting LaTeX code. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

References

- [1] Berkovich, R., Meiran, N., 2023. Pleasant emotional feelings follow one of the most basic psychophysical laws (Weber’s law) as most sensations do. *Emotion* 23(5), 1213–1223.
- [2] Fechner, G.T., 1966. *Elements of Psychophysics*, Volume 1. Holt, Rinehart and Winston, New York. (Original work published 1860).
- [3] Gottman, J.M., Swanson, C., Murray, J., 1999. The mathematics of marital conflict: Dynamic mathematical nonlinear modeling of newlywed marital interaction. *Journal of Family Psychology* 13(1), 3–19.
- [4] Gross, J.J., 1998. The emerging field of emotion regulation: An integrative review. *Review of General Psychology* 2(3), 271–299.
- [5] Kahneman, D., Tversky, A., 1979. Prospect theory: An analysis of decision under risk. *Econometrica* 47(2), 263–290.
- [6] Lewis, M.D., 2005. Bridging emotion theory and neurobiology through dynamic systems modeling. *Behavioral and Brain Sciences* 28(2), 169–187.
- [7] Namboodiri, V.M.K., Mihalas, S., Marton, T.M., Hussain Shuler, M.G., 2014. A temporal basis for Weber’s law in value perception. *Frontiers in Integrative Neuroscience* 8(79), 1–11.
- [8] Thorndike, E.L., 1898. *Animal Intelligence: An Experimental Study of the Associative Processes in Animals*. Psychological Review Monograph Supplements.
- [9] Engel, K.-J., Nagel, R., 2000. *One-Parameter Semigroups for Linear Evolution Equations*. Springer.
- [10] Higham, N.J., 2008. *Functions of Matrices: Theory and Computation*. SIAM.

- [11] Chen, C.-T., 1999. Linear System Theory and Design, 3rd ed. Oxford University Press.
- [12] Ogata, K., 1995. Discrete-Time Control Systems, 2nd ed. Prentice Hall.
- [13] Strogatz, S.H., 2014. Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering, 2nd ed. Westview Press.
- [14] Kuznetsov, Y.A., 2004. Elements of Applied Bifurcation Theory, 3rd ed. Springer.
- [15] Gottman, J. M., Krokoff, L. J., 1989. The relationship between marital interaction and marital satisfaction: A longitudinal view. *Journal of Consulting and Clinical Psychology*, 57(1), 47-52.
- [16] Levenson, R. W., Gottman, J. M., 1983. Marital interaction: Physiological and affective correlates. *Journal of Personality and Social Psychology*, 45(3), 587-597.
- [17] Vallacher, R. R., Nowak, A., 1997. Dynamical social psychology: Finding order in potential chaos. *Psychological Inquiry*, 8(2), 73-99.
- [18] Kloeden, P. E., Platen, E., 1992. Numerical Solution of Stochastic Differential Equations. Springer.



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